

Mohamed Gharieb

Data Scientist | Applied AI Practitioner | Machine Learning Engineer

Alexandria, Egypt · +20 102 248 7565 · m.abdelslam00259@sci.dmu.edu.eg · [LinkedIn](#) · [GitHub](#)

CAREER OBJECTIVE

Results-driven Data Scientist and Applied AI practitioner with a strong academic foundation in Physics and Nanotechnology. Specialized in end-to-end machine learning pipelines—data preprocessing, feature engineering, EDA, model training, and evaluation. IBM & Google certified. Aspire Leaders Fellow (Harvard-founded) and McKinsey Forward Program Graduate. Proven track record of applying data-driven methodologies to real-world problems and leading impactful programs reaching 1,000+ beneficiaries. Passionate about leveraging AI, predictive analytics, and sustainable development (SDGs) for scientific and societal impact.

WORK EXPERIENCE

Applied AI & Data Analytics Trainee · Digilians – Digital Pioneers Initiative (MCIT)

Dec 2025 – Present | Cairo, Egypt

- Selected through a nationally competitive scholarship for Applied AI and Data Analytics led by Egypt's Ministry of Communications & IT (MCIT), in collaboration with top industry and academic partners.
- Build and evaluate supervised ML models (classification, regression) using Python (Scikit-learn, Pandas, NumPy) for predictive analytics on real-world structured datasets.
- Execute full-cycle data preprocessing, feature engineering, and exploratory data analysis (EDA) to extract actionable insights and identify patterns in complex datasets.
- Optimize model performance via hyperparameter tuning, cross-validation, and F1/AUC-ROC evaluation; design data visualizations for both technical and non-technical stakeholders using Matplotlib and Seaborn.

Founder & CEO · LEADUP – Youth-Led Non-Profit Organization

Dec 2022 – Present | Alexandria, Egypt

- Founded and scaled a youth-led non-profit focused on SDG 4 (Quality Education), climate action, and sustainable development, reaching 1,000+ young people across Egyptian governorates.
- Applied data-driven planning and M&E frameworks to measure program impact, optimize outreach, and ensure stakeholder accountability; managed KPI tracking, reporting cycles, and program documentation.
- Led strategic partnerships with UNICEF and the Ministry of Youth; won 3rd Place in the UNICEF Gender & Climate Guidebook competition.
- Delivered training programs on digital skills, climate literacy, and civic leadership to diverse youth cohorts.

Teaching Assistant · Faculty of Science, Damanhour University

Sep 2022 – Jun 2023 | Damanhour, Egypt

- Supported undergraduate Physics courses; guided students through experimental data collection, statistical validation, and scientific reporting.
- Managed LMS course content, prepared assessments, and contributed to accreditation and quality assurance documentation.

PROJECTS

Supervised ML Classification Pipeline (Digilians Capstone)

Tools: Python, Scikit-learn, Pandas, NumPy, Matplotlib, Jupyter Notebook

- Developed an end-to-end ML pipeline for a real-world classification problem: data cleaning, EDA, feature engineering, model training (Random Forest, XGBoost, Logistic Regression), and performance evaluation using F1-score and AUC-ROC metrics; achieved strong generalization via cross-validation and hyperparameter tuning.

Waste-Recycled Nanomaterials for Energy Storage – Published Research

Tools: Literature Analysis, Scientific Writing, Data Synthesis, Nanotechnology

- Investigated sustainable production of carbon nanotubes and graphene from biomass, agricultural, and industrial waste for energy storage and conversion applications; findings published in Nanotechnology Reviews (De Gruyter, 2022) — peer-reviewed international journal.

LEADUP Program Impact Dashboard

Tools: Google Sheets, Tableau, Excel, M&E Frameworks

- Designed a monitoring & evaluation (M&E) data dashboard to track program outreach, beneficiary demographics, and SDG-aligned KPIs across 5 Egyptian governorates, enabling data-driven decision-making and transparent stakeholder reporting.

EDUCATION

Diploma in Nanotechnology · Faculty of Postgraduate Studies for Advanced Sciences, Beni-Suef University
Sep 2024 – Present · Beni-Suef, Egypt

B.Sc. in Physics & Chemistry · Faculty of Science, Damanhour University
Sep 2018 – Jul 2022 · Damanhour, Egypt · GPA: 3.3/4.0 · Graduation Project: Excellent

- Relevant Coursework: Solid-State Physics, Quantum Mechanics, Thermodynamics, Nanotechnology, Analytical Chemistry

CERTIFICATIONS

IBM Data Science Professional Certificate · IBM / Coursera · Dec 2022 – May 2023

- 10-course program covering Python, SQL, machine learning, data visualization, and end-to-end data science project workflows using real-world datasets.

Google Data Analytics Professional Certificate · Google / Coursera · Sep 2021 – Feb 2022

- Comprehensive training in data cleaning, visualization, SQL querying, and R programming for data-driven decision-making.

Machine Learning with Big Data · SRTACity · Jan 2022 – Jun 2022

- Training in ML techniques and big data processing for advanced analytics applications.

FELLOWSHIPS & LEADERSHIP PROGRAMS

Aspire Leaders Fellow · Aspire Institute (Harvard-founded) · Apr – Oct 2025

- Selected through a highly competitive global process; trained in leadership, systems thinking, ethical decision-making, and social impact with an international cohort.

McKinsey Forward Program Graduate · McKinsey & Company · May – Sep 2025

- Developed structured problem-solving, digital transformation, and professional communication skills through McKinsey's globally recognized leadership program.

PUBLICATION

[Cutting-Edge Development in Waste-Recycled Nanomaterials for Energy Storage & Conversion](#) · Nanotechnology Reviews, De Gruyter, 2022

- Investigated sustainable production of carbon nanotubes and graphene from biomass, agricultural, and industrial waste; explored environmental impact of waste recycling for energy storage and conversion in combating climate change.

TECHNICAL SKILLS & KEYWORDS

Programming & Data: Python, Pandas, NumPy, Scikit-learn, Matplotlib, Seaborn, SQL, R, Jupyter Notebook, Git, GitHub

Machine Learning & AI: Supervised Learning, Unsupervised Learning, Feature Engineering, EDA, Predictive Analytics, Hyperparameter Tuning, Cross-Validation, Model Evaluation (F1, AUC-ROC), Computer Vision (Foundational), Generative AI

Visualization & BI: Tableau, Power BI, Microsoft Excel, Google Sheets

Domain Knowledge: Nanotechnology, Solid-State Physics, Quantum Mechanics, Analytical Chemistry, Sustainable Development (SDGs), M&E Frameworks

Soft Skills: Data Storytelling, Strategic Planning, Cross-functional Collaboration, Project Management, Leadership, Stakeholder Engagement

Languages: Arabic (Native), English (C1 – Proficient)